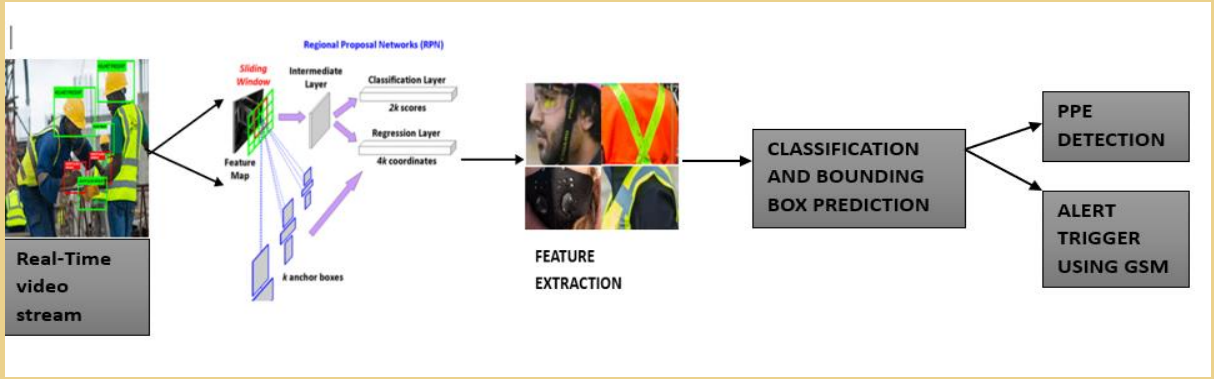
VEL TECH MULTI TECH DR. RANGARAJAN DR. SAKUNTHALA ENGINEERING COLLEGE

AN AUTONOMOUS COLLEGE

CONSTRUCTION SITE SAFETY-ANALYZER AND ENHANCER USING OPTIMIZED YOLOV8 AND R-CNN

ABSTRACT: Our Project introduces an AI-powered system using YOLOv8 and R-CNN to detect PPE compliance on construction sites via live video analysis. It ensures real-time safety enforcement with GSM alerts, reducing manual supervision and enhancing workplace safety.

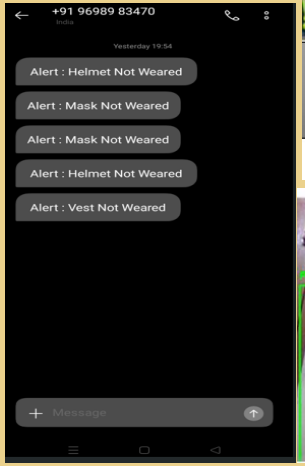


PROBLEM STATEMENT:

How might we develop advanced, non-flammable and lightweight construction materials for the fireworks industry to improve safety, prevent fire hazards, and maintain structural integrity and efficiency in the production environment? (annexure no. 28)

SOLUTION TO OVERCOME:

The system achieved high PPE detection accuracy with YOLOv8, outperforming R-CNN in speed, making it ideal for real-time use. Nomex vests, FRP helmets, masks, and calcium silicate board-built flats, ensured enhanced safety and structural reliability.





CONCLUSION:

The Construction Site Safety Analyzer effectively enhances PPE compliance monitoring using YOLOv8, R-CNN, and GSM-based alert systems. By leveraging computer vision and deep learning, the system automates the detection of helmets, vests, and masks, reducing reliance on manual supervision and minimizing human error.

PROJECT CONTRIBUTORS

MS. P. VINITHA BABY, B.E.,ME.,
ASSISTANT PROFESSOR, PROJECT GUIDE
DEPARTMENT OF AI&DS
+91 73586 18060

R. J. JABIN JOSEPH
TEAM LEADER
DEPARTMENT OF AI&DS
+91 99406 78988

G. BALAMBAL
TEAM MEMBER
DEPARTMENT OF AI&DS
+9163791 33765

L. KISHORE
TEAM MEMBER
DEPARTMENT OF AI&DS
+91 94451 02399

42, Avadi – Vel Tech Road, Poonamallee - Avadi High Rd, Vel Nagar, Chennai, Tamil Nadu 600062. Phone: 073587 01999